CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Banking (Recurring Deposit Accounts)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Interest and maturity value

1. A man deposits ₹500 every month for 18 months in a recurring deposit account and earns ₹570 as interest. Which expression gives the maturity amount? (a) ₹$(500 \times 18 + 570)$ (b) ₹$(500 \times 19 + 570)$ (c) ₹$(500 \times 18 \times 19 + 570)$ (d) ₹$(500 \times 9 \times 19 + 570)$ ICSE 2025, 1 mark

SOLUTION

Maturity amount = total money deposited + interest.

Total deposited $= 500 \times 18$.

Answer: (a)

2. A boy deposits ₹800 every month for 6 months in a recurring deposit account and receives ₹4884 on maturity. How much interest does he earn? (a) ₹84 (b) ₹42 (c) ₹24 (d) ₹284 ICSE 2023, 1 mark

SOLUTION

Total deposited $= 800 \times 6 = 4800$

Interest $= 4884 - 4800 = 84$

Answer: (a)

3. A girl deposits ₹600 a month in a recurring deposit account for $2\frac{1}{2}$ years at 10% per annum. Find the maturity value. ICSE 2018, 3 marks

SOLUTION

$n = 30$ months. Interest $= 600 \times \frac{30 \times 31}{2 \times 12} \times \frac{10}{100} = 600 \times 38.75 \times 0.1 = 2325$

Total deposited $= 600 \times 30 = 18000$

Answer: maturity value = ₹20,325

MathSetu

2 Finding the rate of interest

4. A woman deposits ₹250 a month in a recurring deposit account for 3 years and receives ₹10,110 on maturity. Find (a) the rate of interest (b) how much more interest she would earn by depositing ₹50 more each month, at the same rate and for the same time. ICSE 2025, 4 marks

SOLUTION

(a) $P = 250$, $n = 36$. Total deposited $= 250 \times 36 = 9000$, so the interest is $10110 - 9000 = 1110$.

$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$: $1110 = 250 \times \frac{36 \times 37}{24} \times \frac{r}{100} = 138.75\, r$

$r = \frac{1110}{138.75} = 8$. **The rate is 8% per annum.**

(b) The extra ₹50 a month earns $50 \times \frac{36 \times 37}{24} \times \frac{8}{100} = 50 \times 55.5 \times 0.08 = 222$.

Answer: (a) 8% per annum (b) ₹222 more

5. A man deposits ₹1000 every month in a recurring deposit account for 2 years and receives ₹26,000 on maturity. Find (a) the total interest he earns (b) the rate of interest. ICSE 2023, 4 marks

SOLUTION

(a) Total deposited $= 1000 \times 24 = 24000$, so the interest is $26000 - 24000 = 2000$.

(b) $I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$: $2000 = 1000 \times \frac{24 \times 25}{24} \times \frac{r}{100} = 250r$

$r = 8$

Answer: (a) ₹2000 (b) 8% per annum

6. A man deposits ₹750 a month in a recurring deposit account for 2 years and receives ₹19,125 on maturity. Find the rate of interest. ICSE 2020, 3 marks

SOLUTION

Total deposited $= 750 \times 24 = 18000$, so the interest is $19125 - 18000 = 1125$.

$1125 = 750 \times \frac{24 \times 25}{2 \times 12} \times \frac{r}{100} = 187.5\, r$

$r = 6$

Answer: 6% per annum

3 Finding the time

7. A man deposits ₹2000 a month in a recurring deposit account that pays 8% per annum. He receives ₹1040 as interest on maturity. For how many years was the account held? ICSE 2024, 3 marks

SOLUTION

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} : 1040 = 2000 \times \frac{n(n+1)}{24} \times \frac{8}{100}$$

$1040 = \frac{20}{3} n(n+1)$, so $n(n+1) = 156 = 12 \times 13$

$n = 12$ months

Answer: 1 year

8. A recurring deposit account of ₹1000 a month at 10% per annum earns ₹5550 as interest on maturity. For how long was the account held? ICSE 2018, 3 marks

SOLUTION

$5550 = 1000 \times \frac{n(n+1)}{24} \times \frac{10}{100}$, so $n(n+1) = 1332 = 36 \times 37$

$n = 36$ months

Answer: 3 years

4 Finding the monthly deposit

9. A recurring deposit account is held for 20 months at 9% per annum, and the interest received on maturity is ₹441. Find the amount deposited each month. ICSE 2019, 3 marks

SOLUTION

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} : 441 = P \times \frac{20 \times 21}{24} \times \frac{9}{100} = 1.575P$$

$$P = \frac{441}{1.575} = 280$$

Answer: ₹280 per month

10. A recurring deposit account is held for 3 years at 7.5% per annum, and the interest on maturity is ₹8325. Find (a) the monthly deposit (b) the maturity value. ICSE 2017, 3 marks

SOLUTION

(a) $8325 = P \times \frac{36 \times 37}{2 \times 12} \times \frac{7.5}{100} = P \times 55.5 \times 0.075 = 4.1625P$, so $P = 2000$.

(b) Total deposited $= 2000 \times 36 = 72000$. Maturity value $= 72000 + 8325$.

Answer: (a) ₹2000 (b) ₹80,325

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.